

Can one use an intellectual property (IP) core from OpenCores.org for a commercial purpose? Does he have to pay any fee? What are the conditions governing use of the IP core?

OpenCores advises our users to use the LGPL for their projects, and this licence enables users to use the IP core in commercial projects as well. The GPL can also be used, but it contains a 'grey zone' regarding how it affects hardware, as this license was originally designed for software. No fee or royalty is needed for LGPL and GPL IP cores.

If not monetarily, in what other way can users contribute back to OpenCores?

It is very important that we share the information on how the different projects (IPs) are being used in different designs, as this provides credibility. It is also important that potential bugs are reported back into the Bugtracker system at OpenCores.

At what stages of system design and development do Open IP cores prove to be most handy?

Functionality in hardware is becoming more and more complex, which also means that the verification efforts needed are increasing almost exponentially. So it is very beneficial to release your IP core as Open Source in order to get sufficient verification. Handling all verification internally within a company will become too costly for 90 per cent of the functions—at least for the standard peripheral functions.

At what other stages would Open Sourcing be useful?

Open Source IPs do not "lock you in a corner;" this is very important if you have products with a long lifecycle. If the design is 100 per cent Open Source, it is easy to port the design to other hardware technologies. And, if embedded processors are used, it is possible to guarantee exactly the same software behaviour for a future redesign of the product. And, since the software-related costs in today's products are increasing, this methodology is vital now, so that the engineers can spend time on the next-generation products instead of redesigning old products.

What OpenCores.org is to engineers who are still studying, and what promise does this initiative hold for the electronics industry?

OpenCores serves an important role to engineers who want to learn hardware design. The ability to review many different project source codes (VHDL, Verilog, assembler, C) and test methodologies gives the engineers a faster learning curve than inventing everything on their own.

The electronics industry benefits both from increased competency levels as well as the other benefits that come with using Open Source IPs.

What types of cores (processor, memory, controller, communication, etc) are most abundant on OpenCores.org?

The OpenRISC processor project is very popular and many commercial companies use this processor in their products. And, of course, all peripheral input-output (Ethernet, PCI, USB, SPI, UART) and different memory controller projects are also popular.

Is there any computer-aided software engineering tool recommended by OpenCores.org?

We strongly promote Open Source electronic design automation (EDA) tools like Icarus Verilog simulator, GHDL VHDL simulator and GTKwave waveform viewer.

We feel that it is important to get a complete Open Source EDA tool-chain that allows engineers to design and simulate at the register transfer level, and perform synthesis. It is vital to learn all these in order to become professional engineers in hardware design.

What about tools for circuit simulation, logic simulation, FPGA/ASIC design, etc?

There are more Open Source tools available than people know. We use many of the tools that are, for example, collected in one of Fedora's projects, the Fedora Electronics Lab (<http://spins.fedoraproject.org/fel/#portfolio>).

Are the Open Source IP cores available on OpenCores being used only for verification, etc, or has a chipset been designed and manufactured completely using Open cores?

There are lot many companies using IP cores from OpenCores in their designs for commercial products. The companies are, however, not marketing it actively as they do not want to 'flag' this to their competitors. Linux had the same trend initially until it was noticed as strength and companies started publishing that they used Linux. The same will soon happen with OpenCores.

ORSoc has helped many commercial companies to implement OpenCores IP cores into their commercial products within every industry segment (telecom, multimedia, industrial, space, etc). As I said, 95 per cent of them are not publicising it; that is all.

Is uptake of the Open hardware concept slower than the Open Source software?

The hardware community is much smaller than the software community. This forces us to be more active in order to get feedback and promote the advantages. But we have seen a drastic change after the software community started to accept Linux and other Open Source software systems. Open Source hardware is a couple of years behind software, but I can assure you that we will see the same impressive growth in Open Source hardware in the coming years. ●